

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Chicago, Illinois 60607-7052

Quiz #1

February 5, 2020

Closed Notes and Book

Duration: 20 minutes

The rear window of an automobile is defogged by passing warm air at 40 °C over its inner surface, and the associated convection coefficient is 30 W/(m²·K). Under conditions for which the outside ambient air temperature is -20 °C and the associated convection coefficient is 65 W/(m²·K), what are the inner and outer surface temperatures of the window? The window glass is 4 mm thick and the thermal conductivity of glass is 1.4 W/(m·K).